

## Reasoning:

 This lesson provides students the opportunity to apply their list of good scientific practices to mitigate disease spread in a model town.

  The students work collaboratively to design a controlled experiment to limit the effects of confounding variables and negotiate an appropriate sample size.

 The students will experience how simulations are useful data collection tools when dealing with phenomena that are difficult to control directly.

  Students will critique the methodologies of their peers and utilize the feedback to improve the quality of their experimental design.

  This models how the scientific community provides feedback to one another throughout the scientific process.

| Slide                                                                                                                                                                                                    | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><p>What are your best recommendations of mitigation strategies for the public to stop the spread of this disease and how did you arrive at your conclusions?</p><p>Your Research Question</p></div> | <p> Start the class by reminding the students of the research question.</p> <p> Explain that we are going to distribute the work, just as scientists do, to investigate this as a community.</p> <p> Different groups will focus on different mitigation strategies.</p> |
| <div><p>OBSERVE AND REFLECT !</p></div>                                                                               | <p>Watch until 2:31.</p> <p> Use this video to “set the stage” for the day’s task of mitigating the spread of disease.</p>                                                                                                                                                                                                                                                                                                                 |

## ABOUT THE TOWN

Here's what we know about our town:

- Some parameters have already been set in the simulation, namely:
  - The town cleans surfaces once per day
  - 62% of the population would be willing to get a vaccine
  - 50% of the population would wear masks sometimes under a mandate
- Also, the town refuses to lockdown

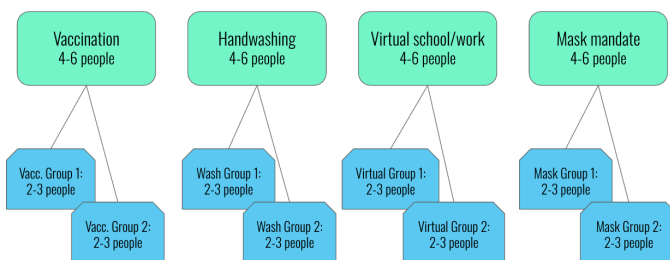
Basically this means that all of the sliders can be left alone, and the lockdown mitigation factor should remain off.

These constraints are present to reduce the complexity of the environment and help students' experiments be more similar, so they are better able to compare and combine results.

 (controlling variables/ maintaining consistency)

## PLANNING YOUR EXPERIMENTAL DESIGN

We'll focus on the remaining mitigation factors:



More groups per factor is also fine, and you could increase the number of students in a group if you'd prefer. However, having at least two groups per factor is highly suggested!

 (how scientific communities build evidence through multiple trials/groups)

## YOUR EXPERIMENT REPORT

What conditions are we interested in?

What's our plan?

Two conditions:

Right now, focusing on:

- Experiment Introduction
- Explanation of Plans
- Condition 1 Method
- Condition 2 Method

| Your mitigation factor: <u>Group 1</u>                                                                                                                                                                                                                                                                                                                                                                      |             | Names: _____                         |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------|--|--------------------|-------------|-------------|--|----------|--|-------------------|--|---------------------|--|---------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------|--|---------|---------|----------|--|-----------|--|----------|--|
| <b>Experiment Introduction</b>                                                                                                                                                                                                                                                                                                                                                                              |             | Why is this important to understand? |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| What specifically are we trying to understand?                                                                                                                                                                                                                                                                                                                                                              |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| <b>Explanation of Plans</b>                                                                                                                                                                                                                                                                                                                                                                                 |             | Why did we plan to do it this way?   |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| What is the plan we made?                                                                                                                                                                                                                                                                                                                                                                                   |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| <table border="1"> <thead> <tr> <th colspan="2">Condition 1 Method</th> </tr> <tr> <th>Mitigation Factors</th> <th># of Trials</th> </tr> </thead> <tbody> <tr> <td>Handwashing</td> <td></td> </tr> <tr> <td>Vaccines</td> <td></td> </tr> <tr> <td>Social Distancing</td> <td></td> </tr> <tr> <td>Virtual Work/School</td> <td></td> </tr> <tr> <td>Mask Mandates</td> <td></td> </tr> </tbody> </table> |             | Condition 1 Method                   |  | Mitigation Factors | # of Trials | Handwashing |  | Vaccines |  | Social Distancing |  | Virtual Work/School |  | Mask Mandates |  | <table border="1"> <thead> <tr> <th colspan="2">Condition 1 Results</th> </tr> <tr> <th>Results</th> <th>Average</th> </tr> </thead> <tbody> <tr> <td>Infected</td> <td></td> </tr> <tr> <td>Recovered</td> <td></td> </tr> <tr> <td>Deceased</td> <td></td> </tr> </tbody> </table> |  | Condition 1 Results |  | Results | Average | Infected |  | Recovered |  | Deceased |  |
| Condition 1 Method                                                                                                                                                                                                                                                                                                                                                                                          |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Mitigation Factors                                                                                                                                                                                                                                                                                                                                                                                          | # of Trials |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Handwashing                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Vaccines                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Social Distancing                                                                                                                                                                                                                                                                                                                                                                                           |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Virtual Work/School                                                                                                                                                                                                                                                                                                                                                                                         |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Mask Mandates                                                                                                                                                                                                                                                                                                                                                                                               |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Condition 1 Results                                                                                                                                                                                                                                                                                                                                                                                         |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Results                                                                                                                                                                                                                                                                                                                                                                                                     | Average     |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Infected                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Recovered                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Deceased                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| <table border="1"> <thead> <tr> <th colspan="2">Condition 2 Method</th> </tr> <tr> <th>Mitigation Factors</th> <th># of Trials</th> </tr> </thead> <tbody> <tr> <td>Handwashing</td> <td></td> </tr> <tr> <td>Vaccines</td> <td></td> </tr> <tr> <td>Social Distancing</td> <td></td> </tr> <tr> <td>Virtual Work/School</td> <td></td> </tr> <tr> <td>Mask Mandates</td> <td></td> </tr> </tbody> </table> |             | Condition 2 Method                   |  | Mitigation Factors | # of Trials | Handwashing |  | Vaccines |  | Social Distancing |  | Virtual Work/School |  | Mask Mandates |  | <table border="1"> <thead> <tr> <th colspan="2">Condition 2 Results</th> </tr> <tr> <th>Results</th> <th>Average</th> </tr> </thead> <tbody> <tr> <td>Infected</td> <td></td> </tr> <tr> <td>Recovered</td> <td></td> </tr> <tr> <td>Deceased</td> <td></td> </tr> </tbody> </table> |  | Condition 2 Results |  | Results | Average | Infected |  | Recovered |  | Deceased |  |
| Condition 2 Method                                                                                                                                                                                                                                                                                                                                                                                          |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Mitigation Factors                                                                                                                                                                                                                                                                                                                                                                                          | # of Trials |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Handwashing                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Vaccines                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Social Distancing                                                                                                                                                                                                                                                                                                                                                                                           |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Virtual Work/School                                                                                                                                                                                                                                                                                                                                                                                         |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Mask Mandates                                                                                                                                                                                                                                                                                                                                                                                               |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Condition 2 Results                                                                                                                                                                                                                                                                                                                                                                                         |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Results                                                                                                                                                                                                                                                                                                                                                                                                     | Average     |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Infected                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Recovered                                                                                                                                                                                                                                                                                                                                                                                                   |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |
| Deceased                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                      |  |                    |             |             |  |          |  |                   |  |                     |  |               |  |                                                                                                                                                                                                                                                                                      |  |                     |  |         |         |          |  |           |  |          |  |

Each group should work on the appropriate tab in the Student Experimentation document.

 During the group work, circulate and take notes on any student comments about the validity of certain choices in methodology. If you do not hear key criteria in the conversation, direct the student to look at the criteria for good scientific models/methods. 

Groups should try to answer both questions in the "Experiment Introduction" and "Explanation of Methods", as well as fill out their plan for both conditions. How they tackle those is up to them.

## Explanations of Each Section of the Experiment Report

- Experiment Introduction
  - Brief notes on what specifically students are investigating, and why. 
  - For instance, are they investigating the effect of vaccines alone, or the effect of vaccines when the town already uses masks and handwashing?
- Explanation of Methods
  - A quick description of how students

plan to run trials, take data, and evaluate their results. 🔍

- Condition Plans
  - Students record the mitigation factors they have turned on/off, as well as the planned # of trials, and what day in the simulation they take measurements at. 🔍
- Condition Results
  - Students record the results of each of their trials here. They don't have to do all six!
- Conclusions from Experiment
  - What do students conclude from these findings?
  - How are these findings insufficient to answering the larger research question? What about considerations not captured in the simulation?  
🧪 (understanding the limitations of evidence collected)

## CRITIQUING OUR EXPERIMENTAL DESIGNS

Two potential ways you can frame this activity:

🧪 🔍 (peer review and consensus building)

1. Students present their designs to the other groups who investigated the same mitigation factor. Remind students the goal is to revise our experiments so we can make stronger inferences about the mitigation factors.
2. Have students critique the designs for homework. Comments can be left in the "Critique" section of the Experiment Report.



Inform students to use the column of peer critique to collect and respond to feedback. Tell students to EXPLAIN why they choose to accept/reject critique when evaluating feedback they receive from their peer groups.

🧪 This helps students see that feedback helps move scientific thinking forward.

### PERFORMING THE EXPERIMENT

Follow your plan for your experiment, and record the results →

Make sure the Condition 1/2 Method sections reflect what you actually did in your experiment

After collecting your results, answer the questions in the Conclusions from Experiment section ↓

| Conclusions from Experiment              |                                           |
|------------------------------------------|-------------------------------------------|
| What can we conclude from these results? | What are the limitations of our findings? |
| <br><br>                                 | <br><br>                                  |

| Condition 1 Results |         |         |         |         |         |         |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|
| Results             | Trial 1 | Trial 2 | Trial 3 | Trial 4 | Trial 5 | Trial 6 | Average |
| Infected            |         |         |         |         |         |         |         |
| Recovered           |         |         |         |         |         |         |         |
| Deceased            |         |         |         |         |         |         |         |

| Condition 2 Results |         |         |         |         |         |         |         |
|---------------------|---------|---------|---------|---------|---------|---------|---------|
| Results             | Trial 1 | Trial 2 | Trial 3 | Trial 4 | Trial 5 | Trial 6 | Average |
| Infected            |         |         |         |         |         |         |         |
| Recovered           |         |         |         |         |         |         |         |
| Deceased            |         |         |         |         |         |         |         |

You could have students create a graph of their data for further analysis.

Student conclusions can be written as homework if class time is limited.

## END OF LESSON 4 REFLECTION

What strategies did we use today that helped improve our experimental designs?  
Why do you think these strategies help?

### Instructions:

Write your thoughts on this in our "Thoughts & Reflections" sheet (in the tab with your name on it).

Point to elevate with the students:



Scientists use procedures that adhere to the norms negotiated within the scientific community as reliable ways to generate accurate claims.



They also continually engage in peer critique to improve their procedures.